

Report: The ArrayQueue Program

In building the `MyArrayQueue` program, my key goal was to make it simple and quick for handling a line. I focused on two main jobs: adding items to the line (enqueueing) and taking items out of the line (dequeuing).

For enqueueing, I made sure the `enqueue` method could easily put items into the line. I set it up so that if the line got full, it would make itself bigger automatically. This meant the line could grow as needed without any trouble.

For dequeuing, I made sure to deal with the case where the line is empty, so we don't try to take out an item that isn't there. This stops any sudden mistakes and keeps the program going well.

I wanted to ensure that both putting in and taking out items stayed quick and easy, no matter if there were a few items or many.

For making it better, I could investigate different ways to make the line bigger to save more space. Instead of making it twice as big every time, I could look for ways to increase it bit by bit to cut down on empty space.

The screenshot included with the report demonstrates how the program works, showing examples of adding and removing items from the queue and how it adjusts its size as needed.

Overall, the `MyArrayQueue` program offers a simple and effective solution for managing a queue using a circular array. But there's always room for improvement, especially in how it handles resizing and potential errors.

```
sarahalrefaie@Sarahs-Laptop java % java MyArrayQueue
[null, null, null]
ENQUEUE
[a, null, null]
ENQUEUE
[a, b, null]
ENQUEUE
[a, b, c]
DEQUEUE
a
[null, b, c]
ENQUEUE
[d, b, c]
ENQUEUE
[b, c, d, e, null, null]
ENQUEUE
[b, c, d, e, f, null]
ENQUEUE
[b, c, d, e, f, g]
DEQUEUE
b
[null, c, d, e, f, g]
DEQUEUE
c
[null, null, d, e, f, g]
ENQUEUE
[h, null, d, e, f, g]
DEQUEUE
d
[h, null, null, e, f, g]
DEQUEUE
e
[h, null, null, null, f, g]
DEQUEUE
f
[h, null, null, null, null, g]
DEQUEUE
g
[h, null, null, null, null, null]
DEQUEUE
h
[null, null, null, null, null, null]
DEQUEUE
Queue is empty
[null, null, null, null, null, null]
sarahalrefaie@Sarahs-Laptop java %
```